Turbulent Boundary Layers in Adverse Pressure Gradients*

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SUMMARY

An experimental study has been made of turbulent boundary layers in adverse pressure gradients which were established in such a way that for each layer the profiles, when plotted in a set of universal coordinates, are of similar shape. A comparison is made with the corresponding set of laminar profiles. Results are presented for the effects of Reynolds Number, pressure gradient, and roughness on skin friction. It is shown that boundary layers in moderately high adverse gradients will not exhibit what is termed downstream stability. A comparison is made of the present experimental results with conventional methods of predicting the effect of pressure gradients on turbulent boundary layers, and it is shown that the agreement is poor.

INTRODUCTION

ONE OF THE MOST CHALLENGING PROBLEMS in fluid mechanics today is that of understanding the behavior of turbulent boundary layers, particularly with regard to their separation or detachment. A great deal of our technology is concerned with devices that are fluid mechanical in nature such as fans, compressors, turbines, pumps, torque converters, reciprocating engines, jet engines, aircraft, missiles, ships, and the like. In almost all cases, the limiting performance of these devices is determined by flow separation. The ultimate lift of a wing or thrust of a propeller is determined by stalling—i.e., separation. Similarly, the maximum compression of an axial-flow fan and the maximum torque of a converter are limited by separation. Even the simplest water faucet depends on separation for its throttling action.

Near every solid surface immersed in a moving fluid there is a layer of fluid retarded by friction. For the fluids that are commonly used, together with the velocities and sizes that most frequently occur, the Reynolds Numbers are sufficiently large that the layers are not only thin but also turbulent. Thus we find that the greater number of our fluid mechanical machines are directly limited in their capabilities by separation of turbulent boundary layers. This separation is caused by the action of adverse or rising pressure gradients, retarding the fluid in the boundary layer, bringing it to rest, and forcing the outer stream to leave the surface and ride out over this stagnant fluid.

In spite of the great technical importance of turbulent separation, our knowledge of the action of pressure gradients on turbulent layers is poor. First, no theory worthy of the name exists for any turbulent shear flow. Second, several empirical methods exist for predicting the effect of pressure gradients on turbulent boundary layers, but experience indicates that none of them is reliable. They do little more than correlate the set of data on which they were based originally.

Early, it was found that the state of affairs in the turbulent boundary layer depends not only upon the local environment such as the local values of the pressure gradient, the wall shear, the surface roughness, the boundary layer thickness, etc., but also upon a long portion of the history of the layer. The forces acting on a boundary layer are surprisingly small. Skin-friction coefficients show that shearing stresses are only a few thousandths of the kinetic energy per unit volume. Furthermore, because the layers are thin, the pressure forces that act upon an effective area across the face of the layer are also very small in size. With such diminutive forces present, the boundary layer reacts slowly to a changing environment. It is for these reasons that the previous history of the layer is so important.

When we began our turbulent boundary-layer research at Johns Hopkins, we wanted to study in detail a few cases that would have simple and well-defined pressure histories and which could be measured with a fair degree of precision. In specifying the history, nondimensional parameters must, of course, be used. The pressure gradient acts across some effective face area δ' of the layer with a force per unit length and width of $\delta'(dp/dx)$. The only other gross force on the layer is the wall shear, τ_0. Consequently, the boundary layer will have a constant history, nondimensionally speaking, if $(\delta'/\tau_0)\ (dp/dx)$ is held constant throughout its length. Since a constant history is a particularly simple and well-defined case, we decided to attempt to obtain such cases, even though we were not sure that it would be easy in practice to set up such "simple" histories. Throughout the rest of this paper we shall, for convenience, call this class of profiles "equilibrium" profiles. This paper describes the results of the work accomplished so far on layers having such profiles.

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Notation

C	=	a constant
c_f	=	local skin-friction coefficient
F	=	power spectrum
F_1, F_2	=	Reynolds Number factors for effect of roughness and pressure gradient on skin friction
f, g	=	functions used in describing velocity profiles
G	=	family parameter of equilibrium profiles
H	=	δ^*/θ
k	=	height of roughness elements
m	=	constant in Falkner-Skan solution
n	=	spectral wave number
p	=	local static pressure
dp/dx	=	static pressure gradient
q	=	local free-stream dynamic pressure
R	=	Reynolds Number
u, v, w	=	components of velocity
U	=	free-stream velocity
u_*	=	friction velocity $= \sqrt{\tau_0/\rho}$
x, y, z	=	position coordinates
$\frac{\Delta u_1}{u_*}$, $\frac{\Delta u_2}{u_*}$	=	shifts of semilogarithmic profiles caused by roughness and pressure gradient
β	=	constant in Hartree solution
δ	=	distance to outer edge of boundary layer
δ^*	=	displacement thickness
δ'	=	effective boundary-layer thickness used in $(\delta'/\tau_0)(dp/dx)$
Δ	=	universal turbulent boundary-layer thickness
θ	=	displacement thickness
ρ	=	fluid density
ν	=	kinematic viscosity
τ_0	=	shearing stress at wall

Experimental Program

Description of Boundary-Layer Tunnel

Our program of turbulent boundary-layer research is a long-range one which has as its goal both the determination of the behavior of turbulent layers and an exploration of their internal structure. A relatively inexpensive wind tunnel was built especially for this research. In establishing conditions for the tunnel design, it was considered necessary that the layers themselves and their associated sublayers should be sufficiently thick that microinstrumentation would not be required. This led to a test section 37 ft. long, 4 ft. high, and 3 ft. wide with velocities ranging from 6 to 40 ft. per sec. Measurements are made on the boundary layer that develops on the floor. The layers range from 2 to 20 in. thick. Because separating boundary layers were to be one of the subjects of investigation, there was great likelihood that adverse gradients would induce separation on the walls and ceiling before it occurred on the floor. This indicated the need for boundary-layer control on the upper three surfaces. To accomplish this without suction ducting and at the same time to simplify the pressure distribution control, the tunnel was arranged as shown in Fig. 1. A large blower is used to force air into the tunnel under pressure. A screened diffusor and a large settling chamber with four single-piece precision screens makes the air stream uniform and of low turbulence. The tunnel is open-return, and a heavily screened diffusor at the exit maintains a controllable overpressure in the working section.

On the sides, simple variable-width slots with rounded edges are provided every 8 in. for boundary-layer removal, the air simply being blown into the atmosphere by the overpressure in the tunnel. Flaps are provided on the top for exhausting relatively large amounts of air to slow down the central stream and establish a variable adverse pressure gradient. A portion of the floor is a flexible masonite sheet, controlled by automobile jacks for more detailed pressure control. A traversing gear for both pitot tubes and hot-wire probes can be used at any location in the test section.

Preliminary Analysis

At first sight it would appear simple to establish a pressure distribution giving a sequence of equilibrium turbulent boundary-layer profiles. Falkner and Skan[1] obtained the theoretical solution for the corresponding set of laminar profiles. The Falkner and Skan solutions show that, if (δ/τ_0) (dp/dx) is held constant throughout the length of the layer, then the profiles of u/U versus y/δ are all identical, even though the Reynolds Number (and, hence, c_f) is changing with x. Further, if x is measured from the origin of the boundary layer, then (x/q) (dp/dx) is likewise constant throughout the length of the layer. This shows that U is proportional to x^m, where $m = -(1/2)$ (x/q) (dp/dx). It is of interest that neither (δ/q) (dp/dx) nor (x/τ_0) (dp/dx) is constant. Since for these laminar layers the profiles are all identical throughout the length of the layer, the choice of thickness in the expression (δ/τ_0) (dp/dx) is arbitrary since all thicknesses will be in constant ratio for a given layer. A simple special case of the above analysis is the Blasius solution, obtained when the pressure is constant.

When we turn to the case of turbulent layers, the situation is not so simple. In contrast to the Blasius case, turbulent profiles are not similar, even when the pressure is constant. At lower Reynolds Numbers, the profiles are roughly one-seventh power profiles, and as the Reynolds Numbers increase, the profiles are more nearly one-eighth or one-ninth power profiles. The typical difference can be seen in Fig. 2 for the profiles of Hama and Klebanoff and Diehl. Further, it is seen that a profile on a rough wall shows even greater lack of similarity.

Normally, one might overlook the deviation for the the smooth plate profiles. However, with our long test section, such discrepancies were likely to show up, and, since we had a chance to make careful measurements under well-defined conditions, we did not want to have effects due to pressure gradients confused with such Reynolds Number effects.

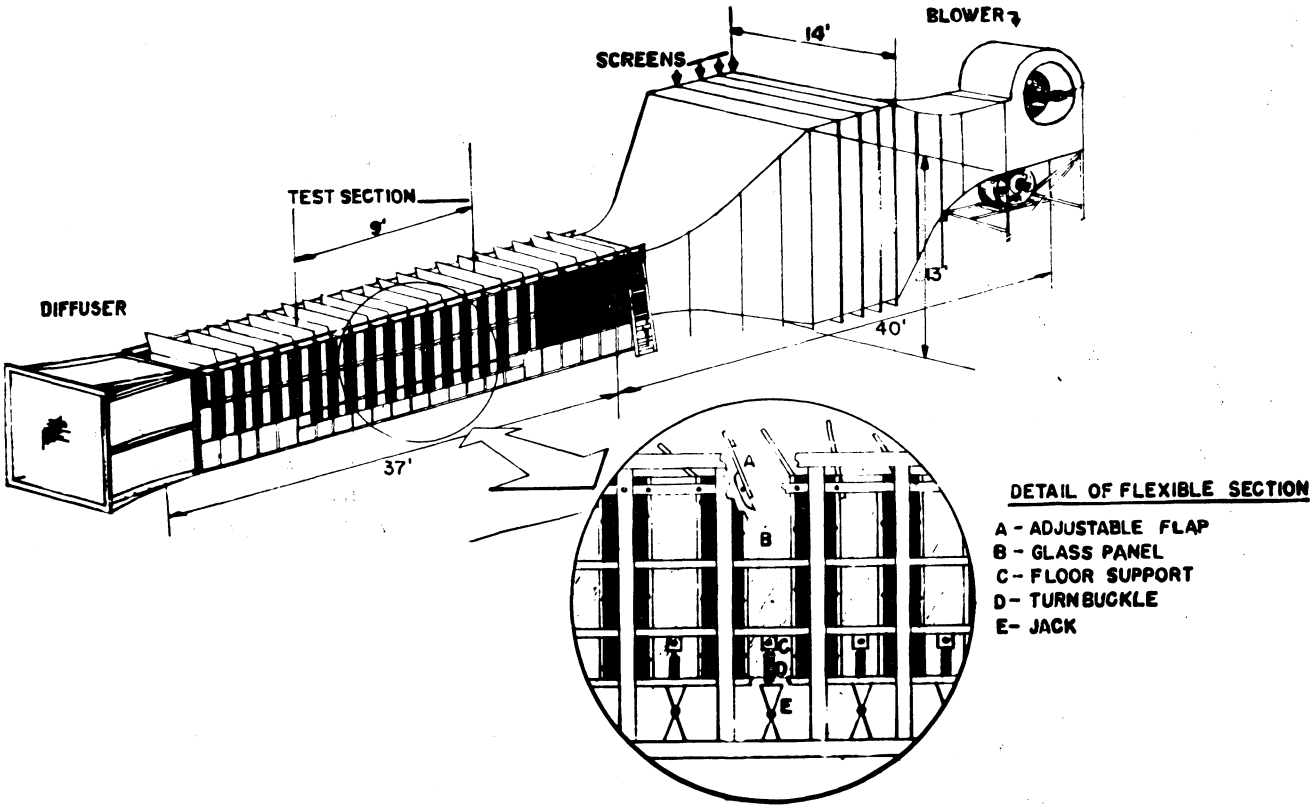

FIG.1 SKETCH OF WIND TUNNEL FOR BOUNDARY LAYER TESTS

Now it has been known for many years that, if we measure velocities relative to the free stream and abandon U as a nondimensionalizing velocity, using in its place $u_* = \sqrt{\tau_0/\rho}$, the friction velocity, then all constant pressure turbulent profiles are similar. In Fig. 3, a collection of profiles has been plotted, and it will be seen they all fall together, including the dissimilar profiles of Fig. 2. This way of plotting retains the typical shape of turbulent profiles. The customary zero velocity point is located at the now variable height, $-\sqrt{2/c_f}$.

It should be emphasized that this plot correlates the profiles independently of both Reynolds Number and roughness. When the roughness size becomes comparable to the thickness of the layer, then the origin of y is undefined. Moore[2] has shown that, even for very large roughness elements, an origin exists which will give agreement with this universal law, as can be seen from his data plotted on Fig. 3. It is believed that all turbulent profiles for which reliable wall shears are known are in agreement with the method of plotting used in Fig. 3. In fact, it is seldom, either in turbulence research or boundary layer research, that a collection of data from varied sources shows such consistency.

The method of correlating turbulent profiles given in the preceding paragraphs offers a possible foundation on which to build an analysis of equilibrium profiles. It appears reasonable that a family of equilibrium profiles should be characterizable by a dimensionless parameter containing the pressure gradient as one of the factors in the parameter. If this is true, then all of the constant pressure profiles must correspond to a single member of the equilibrium family, since for all these profiles the parameter will be zero. This in turn necessitates that a correlation scheme be adopted for which all the constant pressure profiles will be a single member of the family—i.e., they are all identical when plotted nondimensionally. This has been achieved by the method just described.

When we consider equilibrium turbulent layers for which the pressure is not constant, it is necessary to attempt to define more precisely what we mean by equilibrium layers. What we should like, of course, is to have

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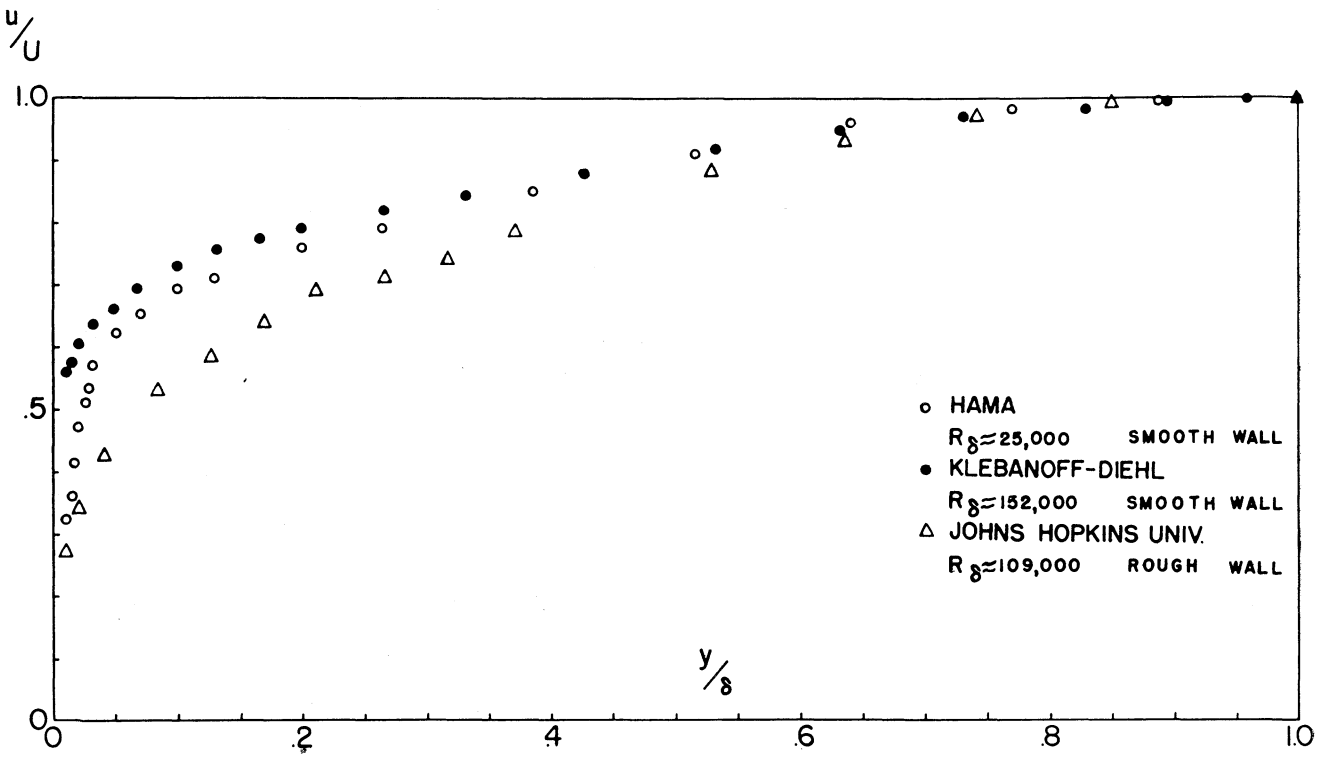

FIG. 2 CONVENTIONAL PLOT OF SEVERAL TURBULENT BOUNDARY LAYER PROFILES FOR CONSTANT PRESSURE

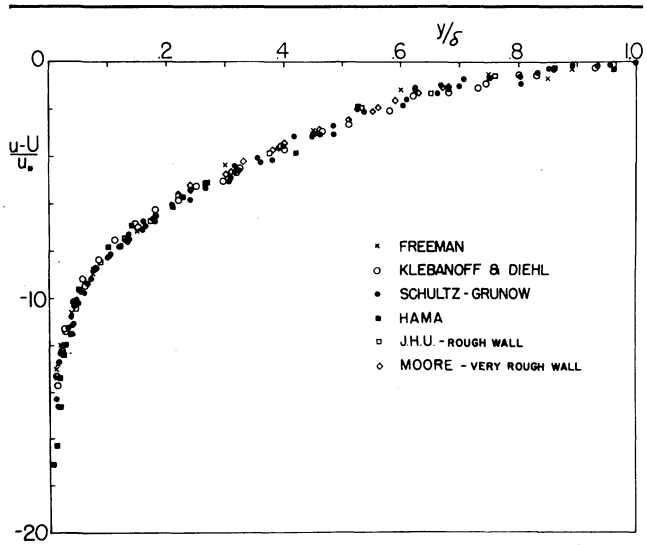

FIG. 3 UNIVERSAL PLOT OF CONSTANT PRESSURE TURBULENT BOUNDARY LAYER PROFILES

a layer for which at every cross section the flow is completely dynamically similar. By this is meant that at each section both the mean velocity patterns and the statistical patterns of the fluctuations will be similar. It is almost certain this cannot be achieved, because as the layer moves down the plate its Reynolds Number increases and this changes the high wave number end of the spectrum compared to the low wave number end. This is clearly seen in the recent measurements of Laufer,[15] which show this change in statistical structure of turbulent shear flow in a pipe when the Reynolds Number is altered.

On the other hand, this same state of affairs must exist in constant pressure layers, and in spite of this an important property such as mean velocity distribution can be correlated on a single plot even when the Reynolds Number is varying. Since large eddies determine the shear distribution, which in turn determines the mean velocities, this probably indicates that for constant pressure layers the small wave number (large eddy size) end of the spectrum is always similar in pattern (i.e., similar as a function of the two variables, wave number, and distance from the wall). The appropriate nondimensional variables would be $F/U_*^2\delta$, $n\delta$, and y/δ.

The foregoing leads to the hope that a like state of affairs may exist for equilibrium layers with varying pressures. If this is true, then it is plausible that equilibrium layers should be those layers for which the profiles at each cross section are identical when $(u - U)/u_*$ is plotted against y/δ. In this report, we shall accept this as a tentative criterion that layers

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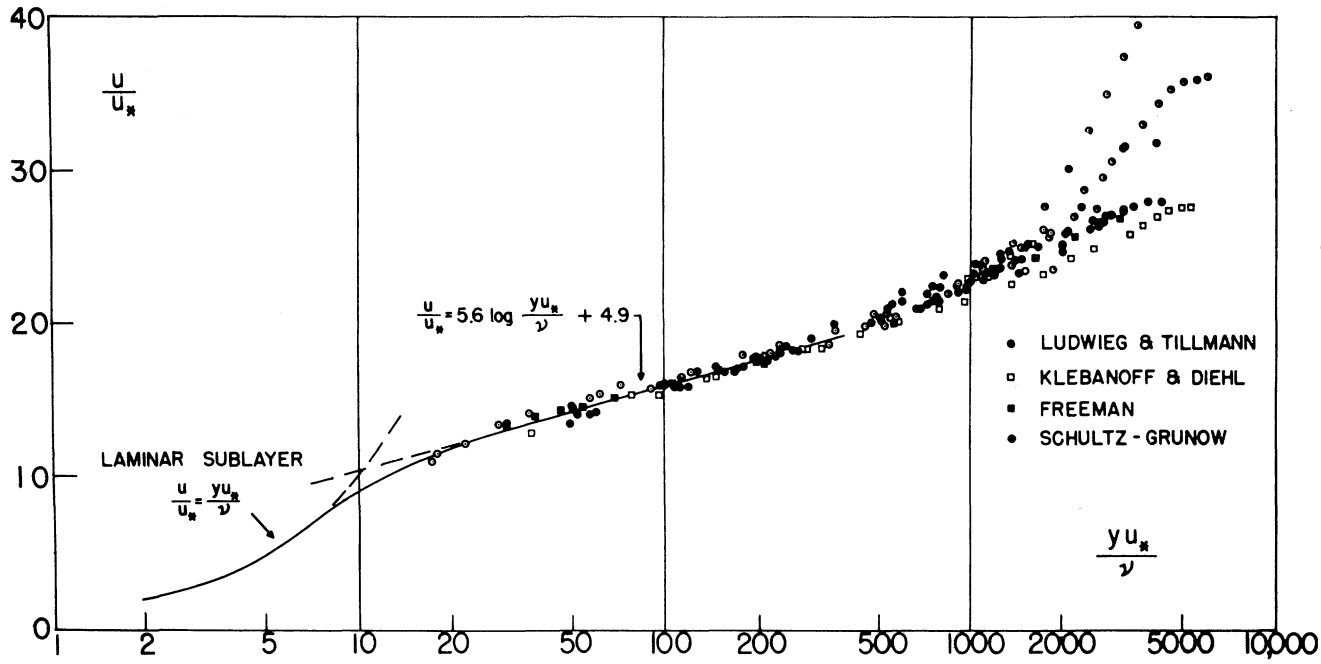

FIG. 4 UNIVERSAL VELOCITY DISTRIBUTION FOR TURBULENT VELOCITY PROFILES NEAR SMOOTH WALLS

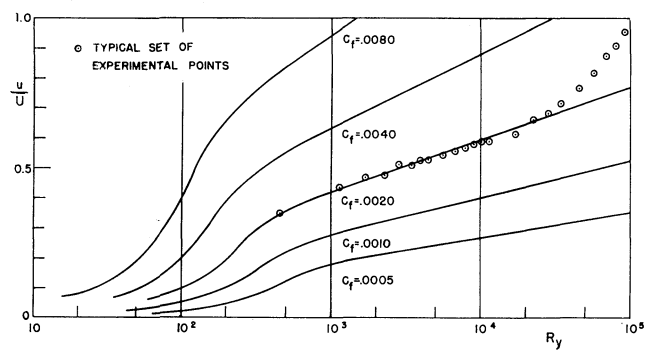

FIG. 5 CHART FOR EXPERIMENTAL DETERMINATION OF TURBULENT SKIN FRICTION

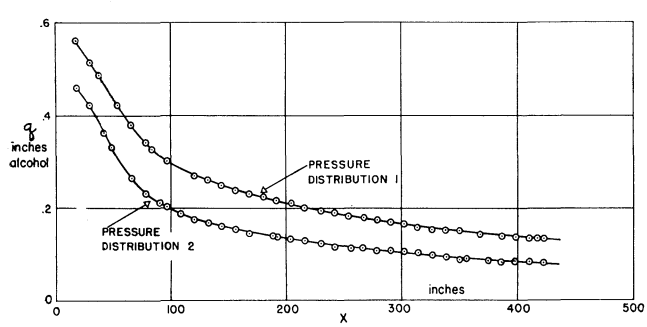

FIG. 6 EXPERIMENTAL PRESSURE DISTRIBUTIONS

are indeed equilibrium layers. Eventually, it will be necessary to show that similarity also exists for some significant set of properties of the fluctuating quantities.

There also remains the question of obtaining a proper parameter of the form $(\delta'/\tau_0)\ (dp/dx)$. We shall show presently that the choice of δ' is not arbitrary as in the case for laminar layers. This point is discussed in some detail later.

Experimental Procedures and Results

It was apparent at the outset of the experimental work that it would be necessary to measure wall skin friction. Not only is τ_0 a useful piece of information in itself but, as has been shown above, it is also a primary datum in the correlation of turbulent profiles. An instrument for measuring τ_0 is being built; however, this instrument was not ready for use when the present results were being obtained so an alternate method was used. Ludwieg and Tillmann[3] found that near a smooth wall, the mean velocity points fall on the well-known universal curve of u/u_* vs. yu_*/ν, even when a pressure gradient is present. A collection of data has been plotted on Fig. 4 to show this agreement. It is sufficiently convincing that the following method of measuring shear has been used: Since $u/u_* = (u/U)\sqrt{2/c_f}$ and $yu_*/\nu = y(U/\nu)\sqrt{c_f/2}$, the universal curve of Fig. 4 can be replotted as a universal family with c_f as a parameter as shown in Fig. 5.† Experimental points taken near the wall are plotted on the figure as shown, and c_f is determined by selecting the appropriate member of the family which fits the points. This method should prove to be of value in turbulent boundary-layer research, since it provides a method of obtaining the skin-friction coefficient simply

† In a private communication, D. Ross has independently proposed to use this same method.

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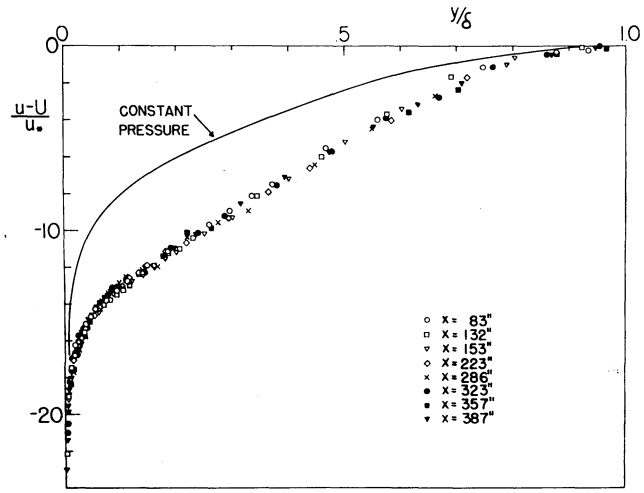

FIG. 7 UNIVERSAL PLOT OF TURBULENT VELOCITY PROFILES FOR PRESSURE DISTRIBUTION I

from the mean velocity profile. This method should make available for analysis a great deal of data taken in the past for which no skin-friction coefficients were obtained but for which accurate velocity profiles were taken.

For the first attempt at establishing an equilibrium set of profiles, it was decided to use a relatively mild dimensionless gradient. This was done because it was known that, with gradients that produce near separating conditions, the velocities close to the wall are small and the turbulence levels large, making accurate measurements difficult because of the lack of existence of instruments having strictly linear response. This initial choice was fortunate for other reasons, as will be seen presently. The procedure adopted was to set up a trial pressure distribution and to measure the velocity profiles at a number of stations, determining the skin-friction coefficients and making a plot of $(u - U)/u_*$ vs. y/δ. For the first trial pressure distribution, the profiles were not similar, and it was necessary to alter the pressure distribution a great number of times before similar profiles were obtained. The final pressure distribution for this first set of measurements is shown in Fig. 6. The corresponding set of similar profiles is shown in Fig. 7. Although the profiles were similar from $x = 83$ to 387 in., the adverse pressure started very near the beginning of the test section. A tripping wire was used at the entrance of the test section to induce a uniform transition. The initial run of the turbulent layer from $x = 0$ to 83 in. was used to give the boundary layer a chance to settle down and attain a reasonably uniform internal dynamical state. The conditions existing during the settling down process have not yet been explored. It is expected this will be done later in the program.

It will be noticed that the profiles of Fig. 7 show an expected difference from those with constant pressure. Compared with this difference the profiles show little scatter from a single curve, indicating that they are nearly identical members of what must be an equilibrium set. The same profiles have been plotted in the conventional manner of u/U vs. y/θ in Fig. 8. It will be seen that there is a consistent variation with increasing x caused by the decrease of c_f as the Reynolds

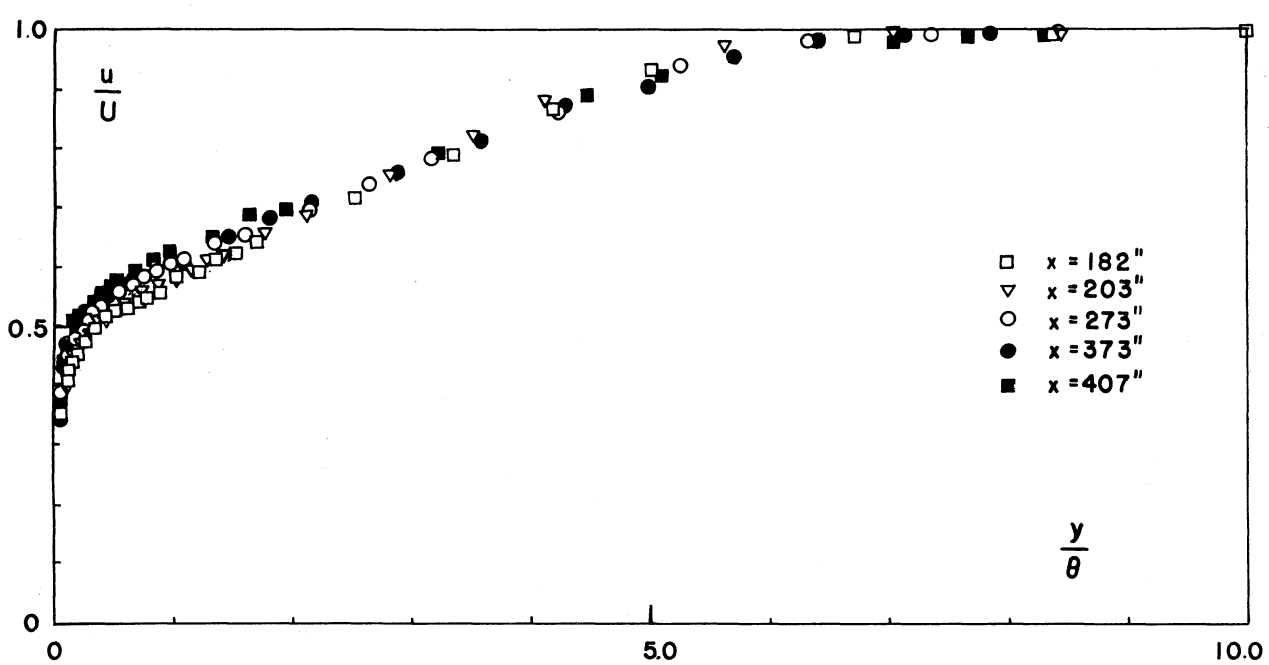

FIG. 8 CONVENTIONAL VELOCITY PROFILES FOR PRESSURE DISTRIBUTION I

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Number increases. This variation of c_f with x is shown in Fig. 9. The change in shape of the profiles on Fig. 8 causes a change in H from about 1.4 to 1.5. Since H for constant pressure profiles is about 1.3 and for separating profiles is about 2.5, it is clear that this set of profiles is far from separation. In spite of this, the skin-friction coefficients are much smaller than corresponding values for constant pressure profiles.

Next it was decided to attempt using a steeper dimensionless gradient. Immediately, serious difficulties were encountered. The first of these was the appearance of strong disagreements with the simple two-dimensional momentum equation,

$$\frac{d\theta}{dx} = \frac{C_f}{2} + \frac{(H+2)}{2}\frac{\theta}{q}\frac{dp}{dx}$$

The c_f term was insignificant, and the dp/dx term was larger than the $d\theta/dx$ term by as much as a factor of two, indicating a lateral divergence of the flow. The second difficulty was an inability to obtain a set of constant dimensionless characteristics for the profiles. Corrections in dp/dx which had previously led to convergence in obtaining equilibrium profiles now produced wild variations at downstream stations. After considerable work these difficulties were understood and corrected. A detailed discussion of the reasons will be given later. The final profiles obtained for the steeper pressure distribution are plotted in Fig. 10. The pressure distribution itself is shown on Fig. 6, along with the first pressure distribution. The skin-friction distribution is also shown on Fig. 9. It is felt that these two sets of constant history profiles, along with the data on constant pressure profiles, give a useful cross section of such nonseparating turbulent profiles in adverse gradients.

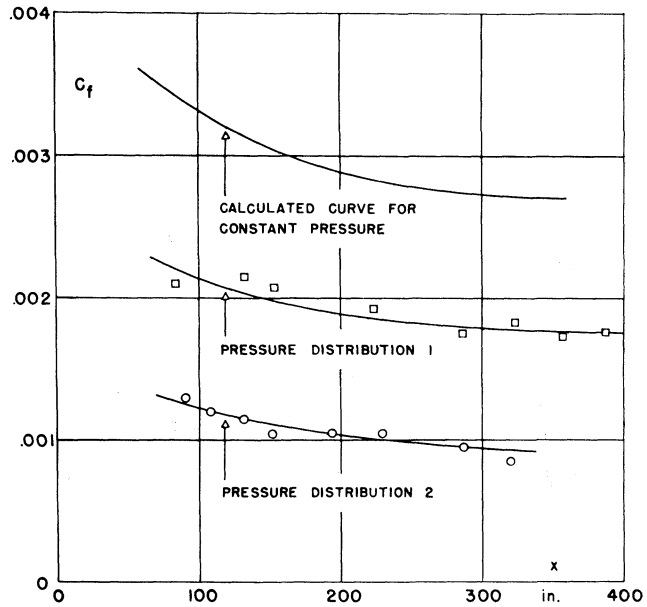

FIG. 9 DISTRIBUTION OF SKIN FRICTION DURING EXPERIMENTS ON TURBULENT VELOCITY PROFILES

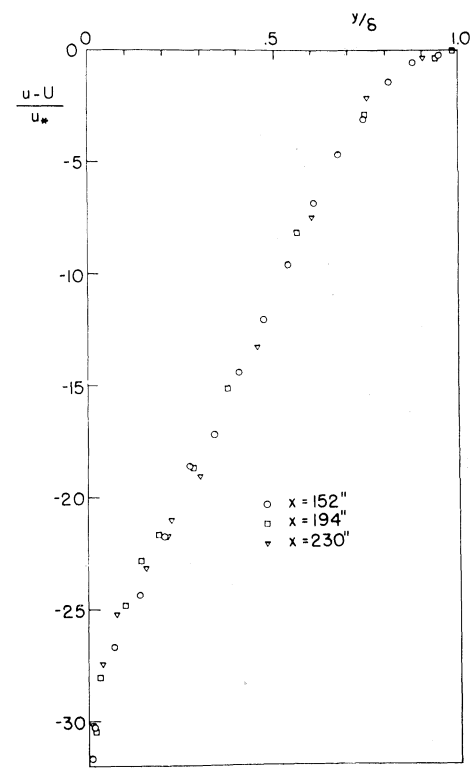

FIG. 10 UNIVERSAL PLOT OF TURBULENT VELOCITY PROFILES FOR PRESSURE DISTRIBUTION 2

Analysis and Discussion

Parameters for Analysis

A question arises as to what parameters can be used to analyze the data that have been obtained. First consider the boundary-layer thickness. The distance δ to the outer edge of the profile has the difficulty that the outer edge of the layer is ill-defined. The displacement thickness

$$\delta^* = \int_0^\infty [1 - (u/U)]\, dy$$

and the momentum thickness

$$\theta = \int_0^\infty [1 - (u/U)]u/U\, dy$$

are not satisfactory for analyzing turbulent profiles because they cannot be obtained directly from the universal plot that forms the basis for our correlation of turbulent profiles. To see this we note that, for each member of the family, $(u - U)/u_*$ is a universal function of y/δ. Thus if we put $(u - U)/u_* = \zeta$ and $y/\delta = \eta$, then $\zeta = f(\eta)$ and we have

$$\delta^* = -\delta\sqrt{\frac{2}{c_f}}\int_0^\infty f(\eta)\, d\eta$$

Since the integral is simply a number, the ratio of δ^* to δ is not fixed, but is a function of c_f—that is, of Rey-

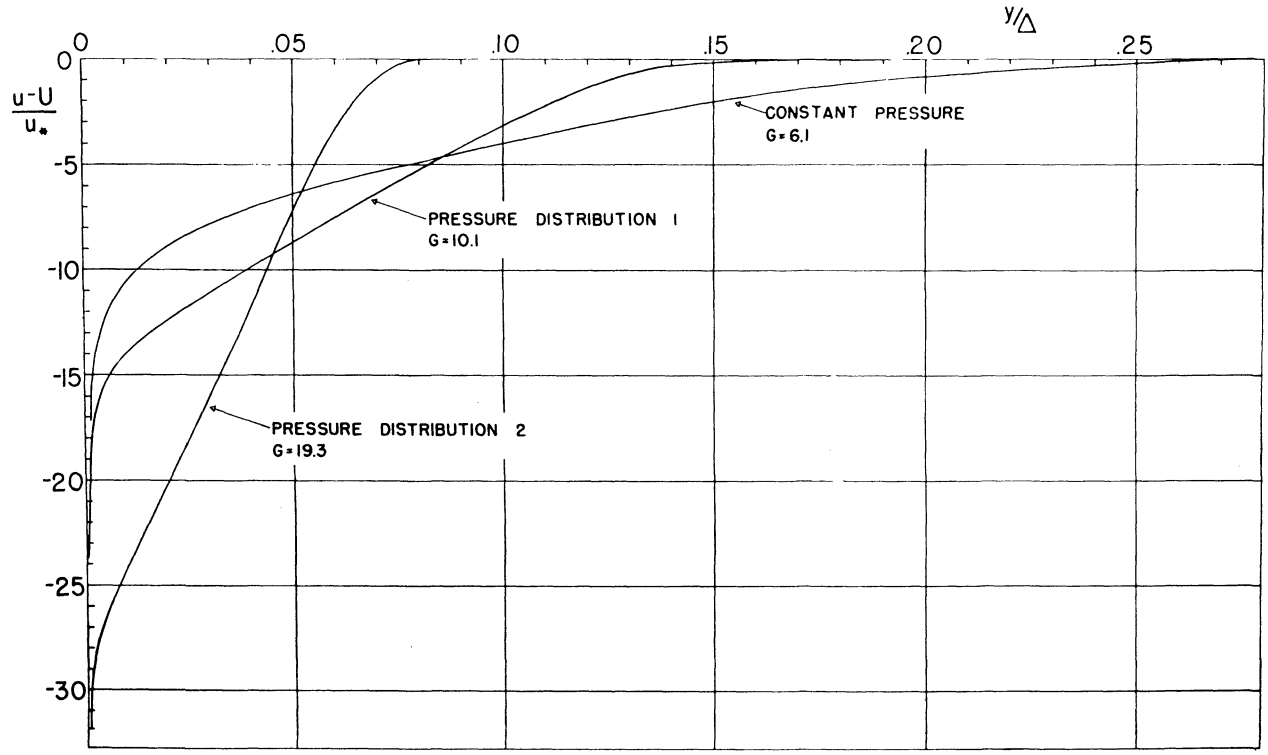

FIG. 11 UNIVERSAL PLOT OF EQUILIBRIUM TURBULENT VELOCITY PROFILES

nolds Number and roughness. A similar statement holds for θ. This dependence of δ^* and θ on Reynolds Number and roughness makes them unsuitable for analyzing a universal family. It is, of course, desirable to have a thickness that does not need these additional data for its specification. It is clear that the simplest universal thickness that is so defined is

$$\Delta = \int_0^\infty \frac{U - u}{u_*} dy = -\delta \int_0^\infty f(\eta)\, d\eta$$

For constant pressure profiles the integral has the value 3.6 Each set of equilibrium profiles will have a different value for the integral, but for each set Δ is in constant ratio to δ. This means that we may substitute Δ for δ in our universal plots and eliminate the uncertainty in determining δ. It will be found that Δ is well suited to analyses of turbulent profiles.

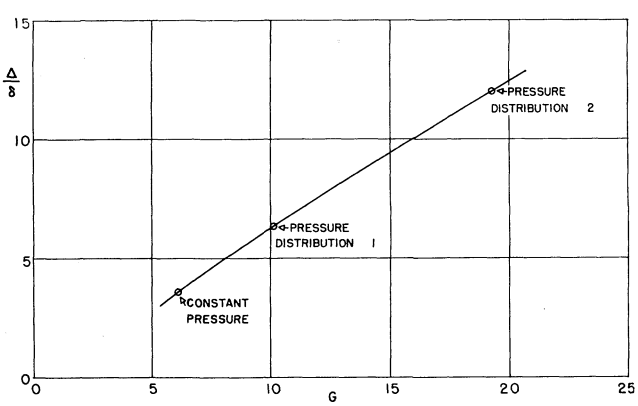

FIG. 12 RELATIONSHIP BETWEEN $\frac{\Delta}{\delta}$ AND G FOR EQUILIBRIUM TURBULENT VELOCITY PROFILES

In addition to Δ, we need a suitable parameter to characterize the various members of our universal family. Customarily, $H = \delta^*/\theta$ is used; however, this suffers from the same shortcoming as δ^* and θ. When profiles are plotted using u/U, δ^* is a first moment of the profile around the axis $u/U = 1$. Similarly, θ is a mixed second moment around the axes $u/U = 0$ and $u/U = 1$, and H is the ratio of these two moments. As a substitute for H, we can ask for a ratio of moments on our universal plot. Since the only axis with significance is $(u - U)/u_* = 0$, the proper choice of parameter is the ratio of the second to the first moment around this axis—i.e.,

$$G = \int_0^\infty \left(\frac{U - u}{u_*}\right)^2 dy \Big/ \int_0^\infty \frac{U - u}{u_*} dy = \int_0^\infty \left(\frac{U - u}{u_*}\right)^2 d(y/\Delta)$$

We can relate our new thickness, Δ, and integral family parameter, G, to θ, δ^*, and H as follows:

$$\delta^* = \sqrt{c_f/2}\ \Delta$$
$$\theta = \sqrt{c_f/2}\,(1 - G\sqrt{c_f/2})\ \Delta$$
$$H = 1/(1 - G\sqrt{c_f/2})$$

In addition to the integral parameter G, we also have the dynamic parameter $(\delta'/\tau_0)\ (dp/dx)$, which, if δ' is properly chosen, will be constant throughout the length of an equilibrium layer. This should, of course, be

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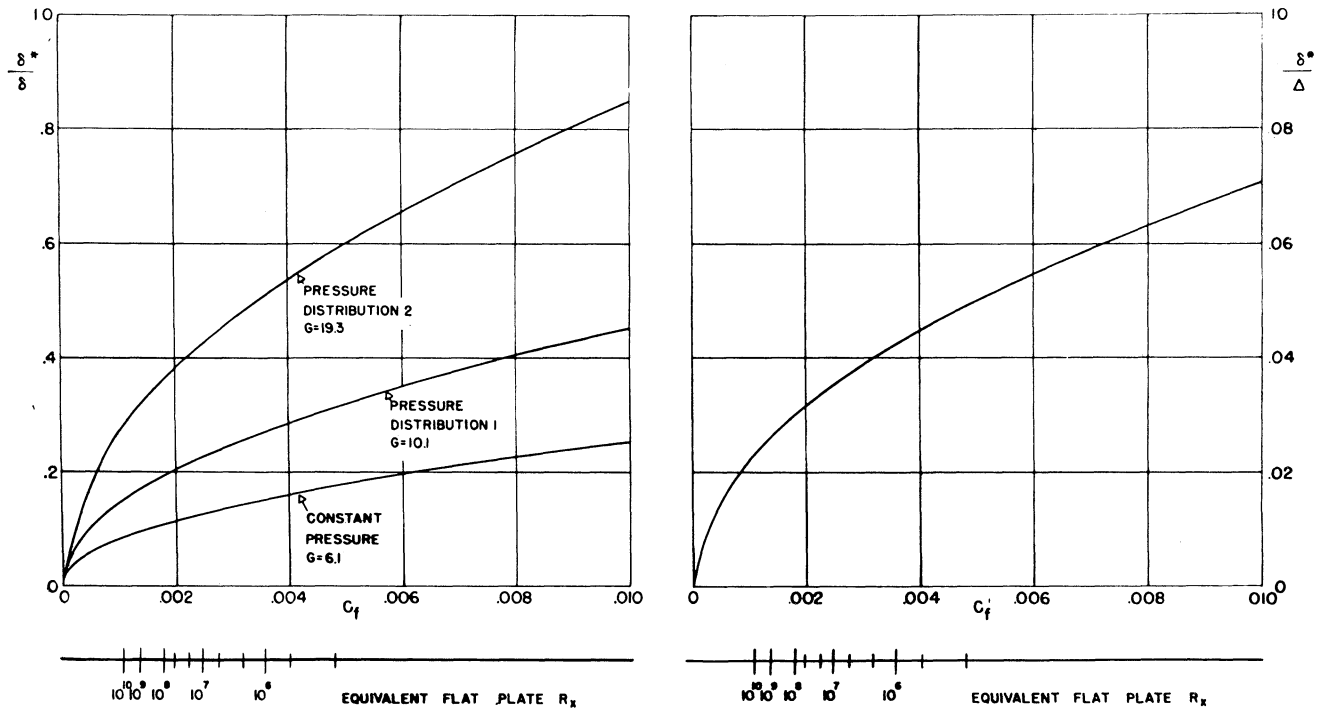

FIG. 13 EFFECT OF C_f ON δ^* FOR EQUILIBRIUM TURBULENT VELOCITY PROFILES

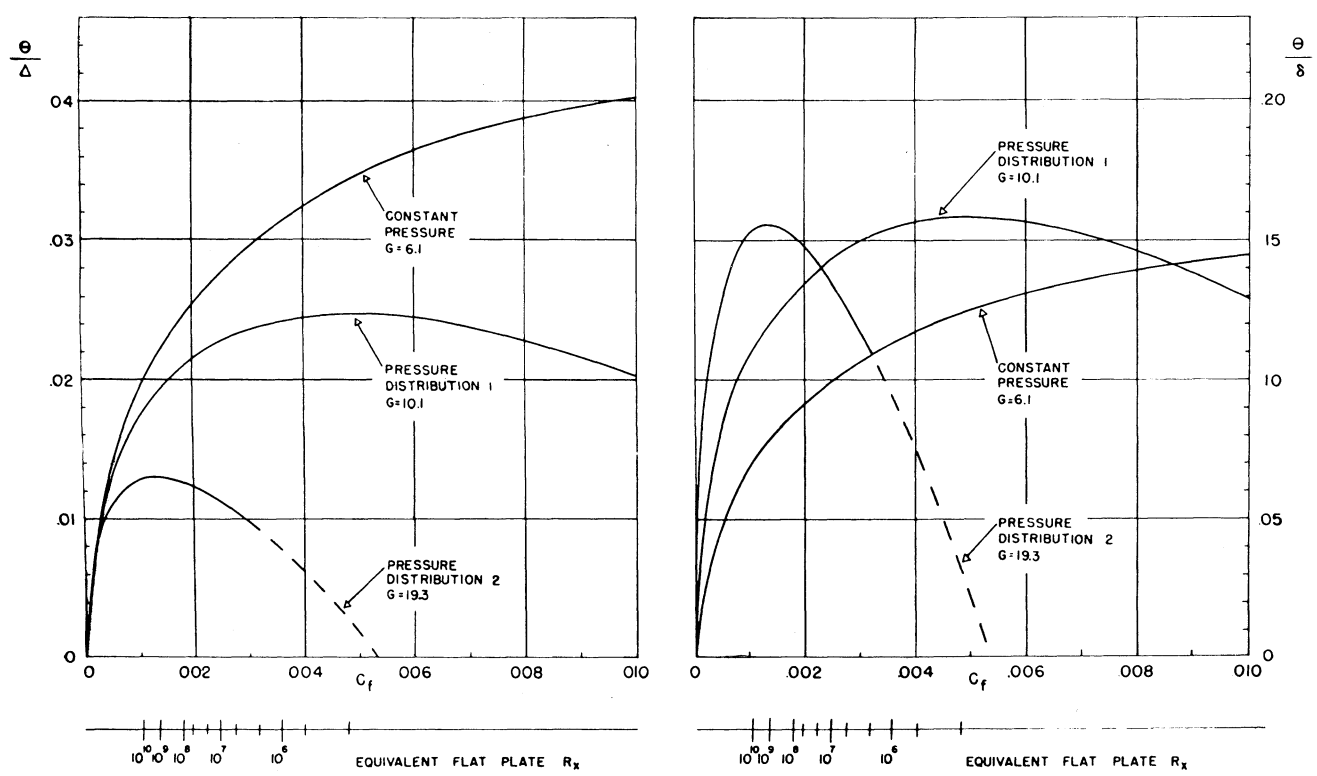

FIG. 14 EFFECT OF C_f ON θ FOR EQUILIBRIUM TURBULENT VELOCITY PROFILES

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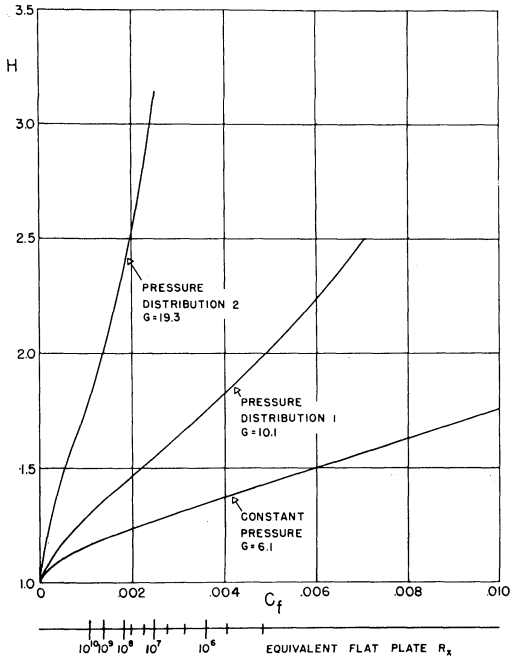

FIG. 15 EFFECT OF C_f ON H FOR EQUILIBRIUM TURBULENT VELOCITY PROFILES

true independently of the value of c_f. However, we have seen that the various thicknesses δ, Δ, θ, and δ^* are not all in constant ratio independent of c_f. Consequently, the choice of thickness to use for δ' is not arbitrary, and we do not know ahead of time which of these thicknesses, if any, is the one that will be appropriate for equilibrium layers. As yet there appears to be no sound theoretical basis for determining which thickness is the proper one. Experimentally this can be obtained fairly simply by establishing equilibrium profiles on both smooth and rough plates, taking care that both have the same values of G, and then finding what choice of δ' will make (δ'/τ_0) (dp/dx) constant. This has not been done as yet.

Analysis of Properties of the Family of Equilibrium Profiles

The data from the two sets of equilibrium profiles, plus the accumulated data for constant pressure profiles, give us three points in plotting the characteristics of the family of equilibrium profiles. A summary of the values found is given in the following table.

Designation	G	Δ/δ
Constant Pressure	6.1	3.6
First Pressure Distribution	10.1	6.4
Second Pressure Distribution	19.3	12.0

In Fig. 11 the profiles have been replotted in terms of the universal parameters given above. The ratio of Δ to δ has been plotted against G in Fig. 12. We also have sufficient information to show the variation of δ^*, θ, and H with c_f. This has been done in Figs. 13, 14, and 15. For convenience, a scale is also given showing the variations of Reynolds Numbers on a smooth flat plate corresponding to the values of c_f. It is apparent that variations of δ^*, θ, and H with c_f are not negligible, showing that an analysis using them as universal parameters would be unsound.

At this point a word of caution is in order. Thus far we have assumed that the entire profile is represented by the universal function of $(u - U)/u_*$ vs. y/δ. This is not true near the wall. On a smooth wall, the profile departs from this universal law in the vicinity of the laminar sublayer. On a rough wall, the departure occurs at a distance from the wall comparable to the height of the roughness. The integrals for Δ and G for a given distribution of $(u - U)/u_*$ vs. y/δ in the outer portion of the layer will have the values quoted above only as long as the laminar sublayer or the roughness elements are small compared with the thickness of the layer. Fortunately this is usually true. This same effect appears in another way. On the profiles of Fig. 11, zero velocity occurs at an ordinate of $-\sqrt{2/c_f}$. As c_f increases, this point occurs at greater values of y/δ on the profile. If this value of y/δ is not small, then substantial portions of the universal profiles will correspond to negative values of u, whereas actual values of u will be positive corresponding to the conditions in the laminar sublayer or in the roughness elements. This will give false values for θ and H. On Figs. 14 and 15, the solid curves have been terminated where this error is no longer negligible. The dotted portions show this incorrect prediction.

Family of Laminar Boundary-Layer Profiles

Since it is of interest to compare our results for the universal family of turbulent boundary layers in adverse pressure gradients with the corresponding results for laminar profiles, we pause at this point to review the properties of the laminar family. We start by asking if a pressure distribution exists such that all the profiles are similar—i.e., that $u/U = f'(y/\delta)$, where δ is some characteristic thickness of the profile and U and δ are functions of x. For convenience of integration, we start with f' as the derivative of a function (stream function) which can be taken so that $f(0) = 0$. We put $\eta = y/\delta$ and insert the expression for u in the continuity equation,

$$(\partial u/\partial x) + (\partial v/\partial y) = 0$$

Remembering that U and δ are functions of x only, we find

$$\frac{v}{U} = \delta' \int_0^\eta \left[\xi f''(\xi) - \frac{\delta U'}{\delta' U} f'(\xi) \right] d\xi$$

where prime denotes differentiation.

Similarity demands that $\delta U'/\delta' U$ be a constant. We can integrate thus

$$\frac{v}{U} = \delta' \left[\eta f'(\eta) - \left(1 + \frac{\delta U'}{\delta' U}\right) f(\eta) \right]$$

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and substitute the result into the boundary-layer equation of motion, giving

$$f'^2 - \left(1 + \frac{\delta' U}{\delta U'}\right) ff'' = 1 + \frac{\nu}{U'\delta^2} f'''$$

This equation was studied by Hartree,[4] who chose the as yet undefined thickness, δ, in such a way that

$$(\nu/U'\delta^2 = 1 + (\delta' U/\delta U')$$

Since $\delta' U/\delta U'$ is constant, $U'\delta^2/\nu$ must be a constant, which Hartree calls β. Falkner and Skan use the constant $m = \beta/(2 - \beta)$. The equation for f becomes

$$\beta(f'^2 - 1) = f''' + ff''$$

The two differential equations for U and δ as functions of x can be integrated explicitly, and, when fitted to the appropriate initial conditions, the solutions are

$$u = Cx^m$$

$$\delta = \sqrt{\frac{2\nu x}{(m+1)U}} = \sqrt{\frac{2\nu}{(m+1)Cx^{m-1}}}$$

Hartree has obtained a family of solutions of the equation for f. Using them, we can compute

$$\delta^* = \int_0^\infty = [1 - (u/U)]\, dy = \delta \int_0^\infty (1 - f')\, d\eta$$

$$\theta = \int_0^\infty u/U[1 - (u/U)]\, dy = \delta \int_0^\infty f'(1 - f')\, d\eta$$

$$c_f = \frac{\tau_0}{\rho(U^2/2)} = \frac{\mu(\partial u/\partial y)_0}{\rho(U^2/2)} = \frac{2\nu}{U\delta} f''(0) = \frac{2f''(0)}{R_{\delta_*}} \int_0^\infty (1 - f')\, d\eta$$

We also see that

$$\frac{\theta}{\tau_0}\frac{dp}{dx} = \frac{\beta}{f'(0)} \int_0^\infty f'(1 - f') d\eta$$

$$\frac{x}{U}\frac{dU}{dx} = -\frac{x}{\rho U^2}\frac{dp}{dx} = m$$

both of which are constant independently of x. We have obtained numerical values for θ/δ, δ^*/δ, H, and (θ/τ_0) (dp/dx) for the Hartree profiles. These profiles are plotted in Fig. 16 with an accompanying legend of useful information.

Effect of Pressure Gradient on Skin Friction for Laminar and Turbulent Profiles

Most methods of treating turbulent boundary layers in adverse pressure gradients assume that the pressure gradients have no effect on the skin friction. Ludwieg and Tillmann[3] and Schubauer and Klebanoff[5] showed that adverse gradients reduced c_f, but neither group was able to make a quantitative statement about the relationship of c_f and dp/dx.

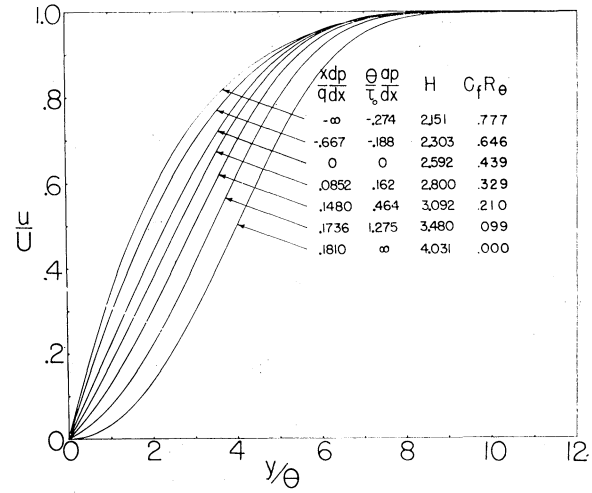

FIG. 16 EQUILIBRIUM LAMINAR VELOCITY PROFILES

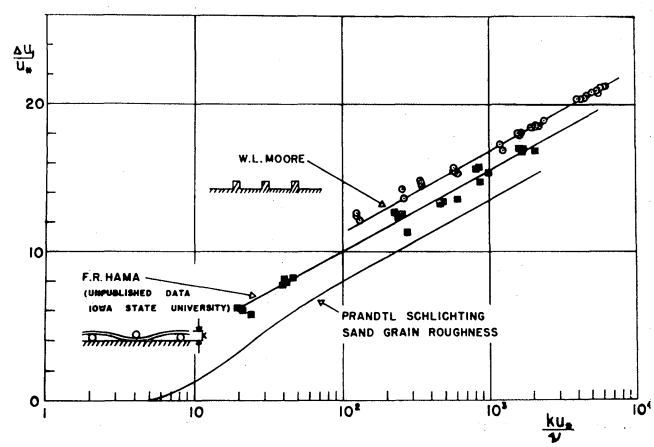

FIG. 17 EFFECT OF ROUGHNESS ON UNIVERSAL TURBULENT VELOCITY PROFILES

Early in the course of the present work it was apparent that pressure gradients had an unexpectedly large effect on skin friction. Even though our first pressure distribution had relatively gentle gradients, the skin-friction coefficients were nearly half of those for constant pressure profiles at the same Reynolds Number.

By combining the results of our present investigation with those of others in the field, we are able to present a relatively complete picture of turbulent skin friction. As mentioned previously, Fig. 4 shows that for smooth plates there is a region outside the laminar sublayer in which the velocity distribution can be represented by the law

$$u/u_* = 5.6 \log_{10} (u_* y/\nu) + 4.9$$

even when pressure gradients are present. Prandtl[6] has shown that, when the wall is rough, the semilogarithmic part of the curve is displaced downward parallel to itself by an amount $\Delta u_1/u_*$, which depends on the roughness Reynolds Number ku_*/ν. In Fig. 17 are

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shown experimental data on the relations between these two parameters for the case of sand grain roughness and also for transverse square bars and wire screening laid on a flat surface. It will be noticed that, when u_*k/ν is greater than approximately 100, the asymptotic form is

$$\Delta u/u_* = 5.6 \log (u_*k/\nu) + \text{constant}$$

Since yu_*/ν at the outer edge of the laminar sublayer is approximately 50, we see that, when the roughness is greater than about twice the thickness of the laminar sublayer, the universal law for the turbulent region near the wall becomes

$$u/u_* = 5.6 \log (y/k) + \text{constant}$$

The above results are known to hold for smooth, moderately rough, and extremely rough walls without pressure gradient and for smooth walls with pressure gradients. Dimensional reasoning, which shows that the effects of pressure gradients on conditions near the wall must be small, indicates that the results are also true for rough walls with pressure gradients.

For all these cases, there exists a turbulent region next to the wall for which u/u_* is linearly related to log y. Turning now to our universal plot of $(u - U)/u_*$ vs. y/Δ, we readily see that, for the inner portion of this plot, $(u - U)/u_*$ should be linearly proportional to log y/Δ. Further, the slope of this semilogarithmic curve should be 5.6—i.e., the same as that on the plot of u/u_* vs. log yu_*/ν or log y/k. This is seen to be true in Fig. 18, where the members of the equilibrium family have been plotted on semilogarithmic scales. The existence of this region of overlapping between the two different plots on which data for the inner and outer portions of the layer may be correlated is of great importance. The overlapping can be used to prove *ab initio* that this portion of the layer must be semilogarithmic,[7] and it serves to establish skin-friction laws for turbulent layers.

On one plot, u/u_* is a universal function of yu_*/ν independent of the free-stream velocity or boundary-layer thickness. The function depends on the surface roughness Reynolds Number u_*k/ν.

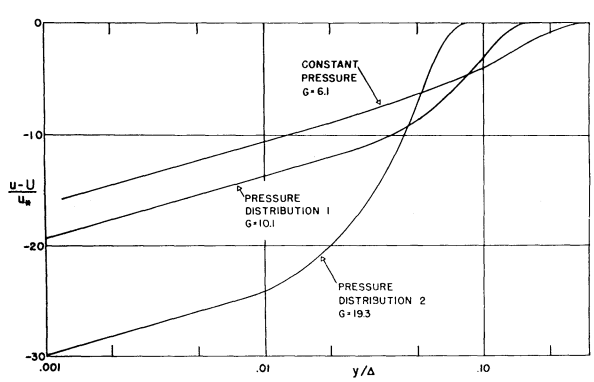

FIG 18 LOGARITHMIC PLOT OF EQUILIBRIUM TURBULENT VELOCITY PROFILES

On the other plot, $(u - U)/u_*$ is a universal function of y/Δ with the function depending on the pressure gradient but being independent of the viscosity (Reynolds Number) or of the roughness. We put

$$u/u_* = g(yu_*/\nu)$$
$$(u - U)/u_* = f(y/\Delta)$$

These two can be rewritten

$$u/u_* = g[(y/\Delta)\cdot(u_*\Delta/\nu)]$$
$$u/u_* = f(y/\Delta) + (U/u_*)$$

Now the experimental data show that there is a common region near the wall where these two quite different relations will simultaneously correlate the velocity profiles. In the overlapping region, both represent the same functional relation between u and y. However, one contains the factor $u_*\Delta/\nu$ as a multiplier of y/Δ inside the function, while the other contains U/u_* as an additive term outside the function. The equivalence of the two equations is maintained even though $u_*\Delta/\nu$ and U/u_* vary through a wide range of values. This implies both that a relation exists between $u_*\Delta/\nu$ and U/u_* and that the functions f and g must be logarithms, since this is the only function for which a multiplicative factor inside the function has the same effect as an additive factor outside the function. This, of course, is in agreement with the experimental data that show that the functions are indeed logarithmic.

The function g can be written

$$u/u_* = 5.6 \log_{10} yu_*/\nu - (\Delta u_1/u_*) + 4.9$$

and the function f can be written

$$(u - U)/u_* = 5.6 \log (y/\Delta) - (\Delta u_2/u_*) + 0.6$$

for those portions of the curves on Fig. 18 that are semilogarithmic. The function $\Delta u_1/u_*$ is the shift in vertical ordinate of the straight line on Fig. 4 when roughness is present. It is a function of the variable ku_*/ν as shown on Fig. 17. The function $\Delta u_2/u_*$ is the corresponding shift of the straight-line portion of the curves on Fig. 18. The shift is measured from the line of the constant pressure profile. It is a function of the pressure gradient, G, or equivalently of the parameter $(\delta'/\tau_0)\ (dp/dx)$.

The relationship between U/u_* and $u_*\Delta/\nu$, mentioned above, can be obtained by subtracting the above equations, which eliminates both u and y; thus,

$$U/u_* = 5.6 \log (u_*\Delta/\nu) - (\Delta u_1/u_*) + (\Delta u_2/u_*) + 4.3$$

or since $u_*/U = \sqrt{c_f/2}$ and $\sqrt{c_f/2}\Delta = \delta^*$,

$$\sqrt{\frac{2}{c_f}} = 5.6 \log R_{\delta_*} - \frac{\Delta u_1}{u_*}\left(R_k\sqrt{\frac{c_f}{2}}\right) + \frac{\Delta u_2}{u_*}(G) + 4.3$$

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This equation is of remarkable significance. It is the universal skin-friction law for all turbulent boundary layers having equilibrium pressure gradients. It includes the effects of Reynolds Number, pressure gradient, and surface roughness. It is probably the most general "universal" law that exists, because other pressure distributions will have an arbitrariness of definition that will make the appelation "universal" inappropriate.

This law contains the four parameters c_f, R_{δ_*}, G, and R_k. R_{δ_*} may be replaced by R_θ and H by G from the following formulas:

$$R_\theta = (1 - G\sqrt{c_f/2})R_{\delta_*}$$

$$H = 1/(1 - G\sqrt{c_f/2})$$

The existence of four parameters makes the skin-friction law difficult to plot for practical use. In Fig. 19, a useful set of curves is given for the case of smooth plates where $k = 0$. Lines are shown for both G constant and H constant. Although H is frequently more convenient to use, a given equilibrium boundary layer will move along a G curve in its development. Although it is cumbersome to plot the four-parameter skin-friction law, it can fortunately be put in a form for ready engineering application. If we introduce two factors F_1 and F_2, the skin-friction law can be written

$$\sqrt{2/C_f} = 5.6 \log [R_{\delta_*}F_2/F_1] + 4.3$$

When F_1 and F_2 are unity, this is the well-known skin-friction law for smooth plates with constant pressure, which has been plotted in Fig. 20. The factors

$$F_1 = 10^{\Delta u_1/5.6u_*}, \qquad F_2 = 10^{\Delta u_2/5.6u_*}$$

have been plotted in Figs. 21 and 22, using the data of the present report for pressure gradients and the Prandtl-Schlichting data for sand grain roughness. In use, after F_1 and F_2 have been obtained from Figs. 21 and 22, a fictitious Reynolds Number $R_{\delta_*}F_2/F_1$ is computed. This is used in place of the actual Reynolds Number, R_{δ_*}, to find c_f in Fig. 20.

Referring back to the analysis for laminar profiles, the formula

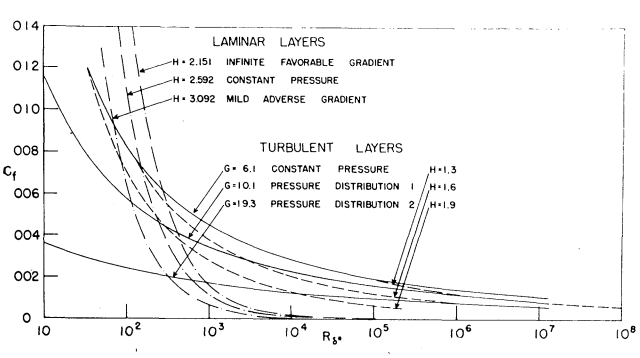

FIG. 19 EFFECT OF PRESSURE GRADIENTS ON LAMINAR AND TURBULENT SKIN FRICTION FOR SMOOTH PLATES

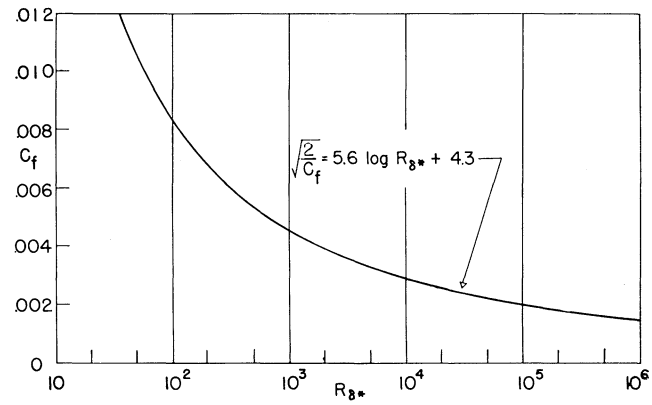

FIG. 20 CONVENTIONAL TURBULENT SKIN FRICTION LAW FOR SMOOTH PLATES WITH CONSTANT PRESSURE

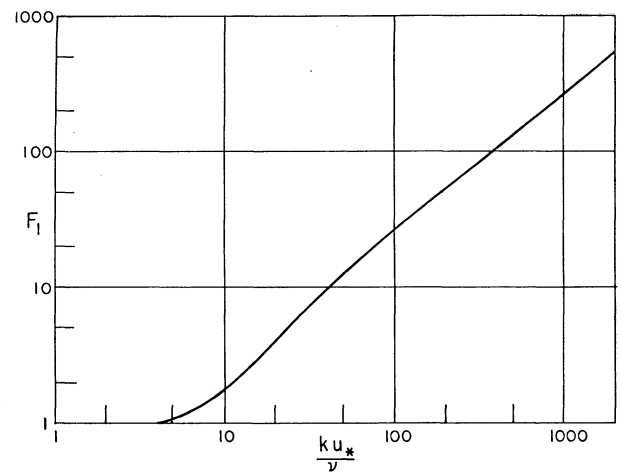

FIG. 21 REYNOLDS' NUMBER FACTOR FOR THE EFFECT OF SAND PAPER ROUGHNESS ON SKIN FRICTION

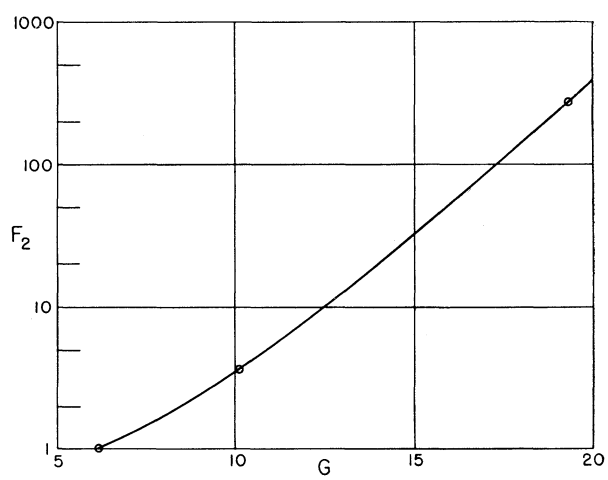

FIG. 22 REYNOLDS' NUMBER FACTOR FOR THE EFFECT OF PRESSURE GRADIENT ON SKIN FRICTION

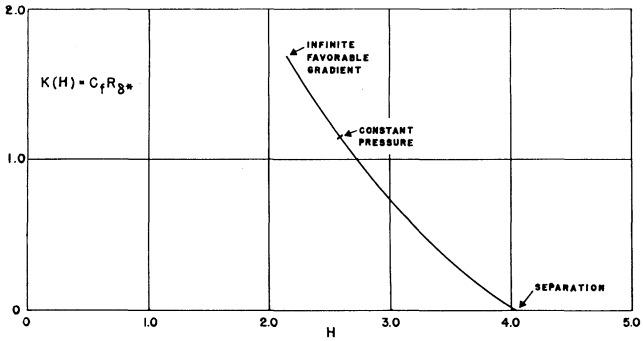

FIG. 23 EFFECT OF H ON LAMINAR SKIN FRICTION COEFFICIENTS

$$c_f = \frac{2f''(0)}{R_{\delta*}} \int_0^\infty (1 - f')\, d\eta = \frac{K(H)}{R_{\delta*}}$$

was obtained. The factor $K(H)$ has been evaluated and is plotted in Fig. 23. Laminar skin-friction coefficients have been plotted in Fig. 19, along with the data for turbulent layers. It will be noticed that the effects of pressure gradient on skin friction are large, both for laminar and turbulent layers. Gradients considerably smaller than those required to produce separation are effective in reducing the surface friction. It is felt that this reduction is of great technical importance.

Comparison of Present and Past Results

It is of interest to give a comparison of the results of the present paper with those of previous authors. Since an understanding of turbulent processes under the combined influences of shear and pressure gradients does not exist, analyses have necessarily used the momentum integral equation,

$$\frac{d\theta}{dx} = (c_f/2) + \frac{(H+2)}{2}\frac{\theta}{q}\frac{dp}{dx}$$

as a starting point. Since the momentum equation is not complete in itself, it is necessary to make assumptions about how the terms in the equation are to be obtained. These assumptions may generally be cataloged as follows:

(1) It is assumed that only a single family of possible turbulent profiles exists. This permits the necessary relationship between δ^* and θ to be determined. Different authors have used different parameters to characterize the various members of the family of profiles.

(2) It is necessary to introduce an auxiliary equation to determine the successive values of the family parameter. This auxiliary equation is invariably based on empirical correlations of collections of experimental data.

(3) A law for the variation of skin friction must be assumed. Most authors have assumed a power-law variation with Reynolds Number, neglecting any influence of pressure gradient on skin friction.

In making the comparison we shall follow the above three points. We shall deal principally with the method of analysis proposed by von Doenhoff and Tetervin,[8] since this method is one of the better known in this country and is generally typical of the rest.

Von Doenhoff uses a universal family of profiles in which u/U is plotted against y/θ. The family parameter is H. Since all the quantities entering into this formulation may be obtained directly from a velocity profile, it is an easy matter to check to see if any given profile fits the family. This has been done for typical profiles from our first and second pressure distributions, and the results have been plotted on Fig. 24. The great and consistent difference between the two sets of profiles, which should agree, was quite unexpected. It had been felt that this single-parameter family assumption was one of the relatively reliable points of prior analyses. The difference shown is sufficient to cast serious doubt upon this assumption for general boundary-layer behavior. The difference is probably caused by the fact that in our profiles H was nearly constant whereas, in von Doenhoff's profiles, H was increasing fairly rapidly. Apparently this has an effect on the shape of the profile.

Von Doenhoff uses, as an auxiliary equation,

$$\theta \frac{dH}{dx} = e^{4.68(H-2.975)} \left[2\frac{\theta}{\tau_0}\frac{dp}{dx} - 2.035\,(H - 1.286) \right]$$

In analyzing his data, he assumes that the base profile for zero pressure gradient is a one-seventh power profile for which $H = 1.286$. As explained earlier, this is not in agreement with our knowledge of constant pressure profiles as shown in Figs. 2 and 3. Actually, H varies with c_f even for constant pressure, as shown in Fig. 15. If the range of Reynolds Number is small and only smooth plates are considered, the variation of H with c_f is not large, and it can be argued that neglecting the variation is not serious. Consistently with this, one may follow von Doenhoff in viewing any H = constant set of profiles as analogous to what we have been calling equilibrium profiles. The reasoning here is analogous to that given later in connection with Fig. 26. For H = constant, von Doenhoff's equation becomes

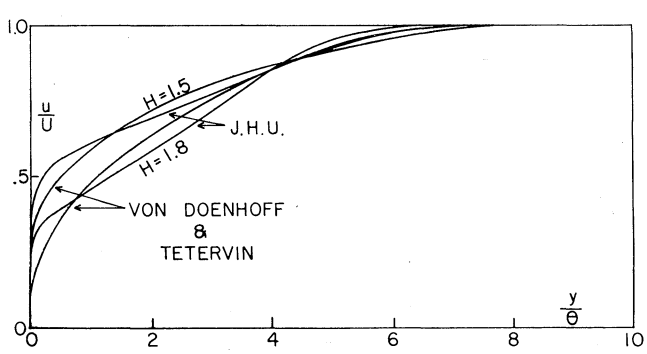

FIG. 24 COMPARISON OF EQUILIBRIUM PROFILES AND THE VON DOENHOFF - TETERVIN ONE PARAMETER FAMILY OF PROFILES

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$$(\theta/\tau_0)\ (dp/dx) - 1.018\ (H - 1.286)$$

This can be compared directly with our data, which have been plotted as a band on Fig. 25 along with von Doenhoff's line. The results are quite discrepant. The only agreement is for $(\theta/\tau_0)\ (dp/dx) = 0$—i.e., for constant pressure profiles where they must agree reasonably well. After that, the trends are quite different. Von Doenhoff's equation predicts a small value of $(\theta/\tau_0)\ (dp/dx)$ at separation, whereas in actuality it must be infinite because τ_0 goes to zero. His linear relation misses completely the bending over of the experimental data. As will be explained in more detail in connection with Fig. 26, the reason the present data are plotted as a band is because the parameters H and $(\theta/\tau_0)\ (dp/dx)$ are not consistent with the universal parameters we find appropriate for turbulent layers. The scatter is caused by the variation of H and θ with c_f.

For skin friction, von Doenhoff uses a power-law variation with Reynolds Number and assumes that there is no effect of pressure gradient. As has been seen, the present results show that the pressure gradient effect is large.

In summary, it is seen that the experimental results we have obtained show little or no agreement with the predictions of the von Doenhoff method. The same is true of other methods with which the author is familiar. Since these methods do not agree with the relatively simple and well-defined experimental results we have obtained, one can conclude that the field is still wide open for the advent of a reliable method of predicting the behavior of turbulent layers under the influence of pressure gradients.

Downstream Stability

It will be recalled that difficulty was experienced in obtaining constant values of G for the steeper set of pressure gradients. Throughout most of turbulent boundary-layer work there exists a fundamental proposition that is seldom enunciated explicitly but which is generally believed to be true. It states that a turbulent boundary layer has a downstream stability such that in a given environment the effect of any small perturbation, such as local changes in roughness or pressure distribution, will die out downstream, leaving no trace other than a possible slight thickening of the layer. Thus, the effect of a crack or a wave in the wall will have no significant effects on the structure of the boundary layer far downstream. Klebanoff and Diehl[9] found, indeed, that constant pressure layers can be violently disturbed and yet return to a universal state downstream, several times thicker but otherwise none the worse for their rough treatment. Similarly, for our experiments with the flatter pressure distribution, the layer possessed this downstream stability, permitting corrections to be made at one point without unduly disturbing downstream points.

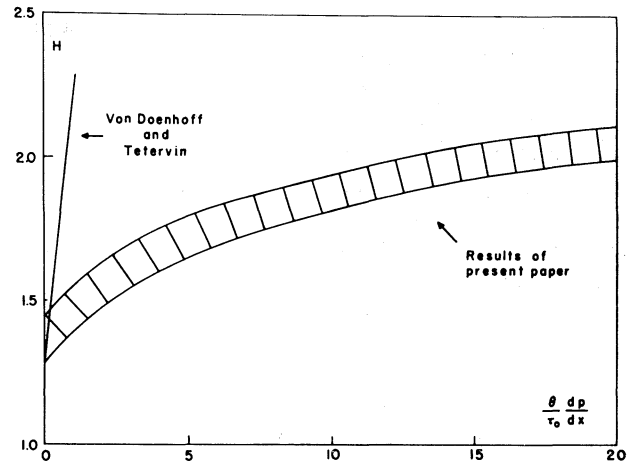

FIG. 25 COMPARISON OF PRESENT AND PREVIOUS RESULTS FOR H versus $\frac{\theta}{\tau_0}\frac{dp}{dx}$ FOR EQUILIBRIUM TURBULENT LAYERS

When work was first started on the steeper gradients, it was not realized that the situation might be different. Corrections were made simultaneously at several points, and it quickly became apparent that something was wrong. It will be seen that downstream stability is necessary to ensure that such simultaneous changes will lead to a converging result. Since we were sufficiently far from separation that upstream influences were not present, the chaotic response that occurred when this procedure was followed indicated that some kind of downstream instability might be present. The following analysis shows why this was so.

Intuitively, one believes that $(\theta/\tau_0)\ (dp/dx)$ should be a reasonably well-behaved dynamic parameter. For constant pressure profiles it is zero. For separating profiles it is ∞, because τ_0 vanishes at separation. In between, it should provide a reasonably reliable indication of the state of the layer. Particularly for equilibrium profiles, it should monotonically increase as one proceeds from constant pressure profiles to separating profiles. As explained previously, its value for turbulent profiles will be somewhat dependent on c_f, but this will not affect the present analysis. The

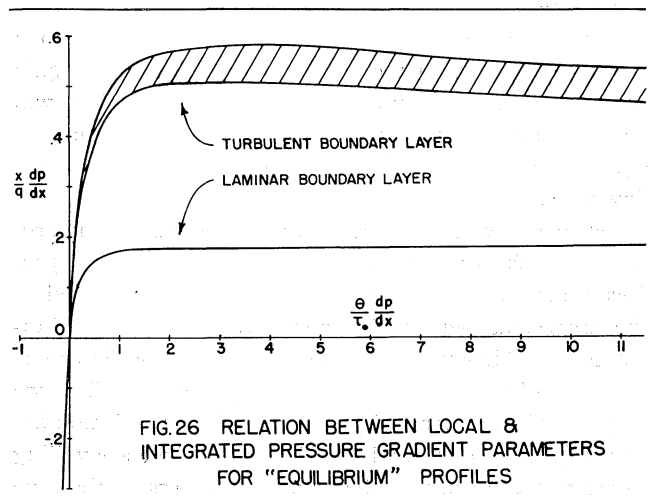

FIG. 26 RELATION BETWEEN LOCAL & INTEGRATED PRESSURE GRADIENT PARAMETERS FOR "EQUILIBRIUM" PROFILES

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difficulty in its use is that when one is setting up a pressure distribution, either for a boundary-layer test or for the more general problem of the design of fluid mechanical machinery, one deals not with θ and τ_0 but with x and p or q. Correspondingly, one would like to use, instead of $(\theta/\tau_0)\ (dp/dx)$, dimensionless parameters such as $(x/q)\ (dp/dx)$ or $(x/U)\ (dU/dx)$. For specific families such as the equilibrium profiles, we can ask for the relationship between the two types of parameters.

To illustrate, consider the equilibrium laminar profiles. We have seen previously that for these profiles $(x/q)\ (dp/dx)$ and $(\theta/\tau_0)\ (dp/dx)$ are both constant throughout the length of the layer. Consequently, for equilibrium layers there will be an exact relation between these parameters. This is shown in Fig. 26. Near zero, $(x/q)\ (dp/dx)$ increases proportionately to $(\theta/\tau_0)\ (dp/dx)$; however, when $(\theta/\tau_0)\ (dp/dx)$ is greater than 1, $(x/q)\ (dp/dx)$ is nearly constant at its asymptotic value of 0.181. This means that, if one were to attempt to obtain profiles having a value of $(\theta/\tau_0)\ (dp/dx)$ greater than approximately unity, the pressure gradient would have to be set with an almost impossible precision to avoid getting a different $(\theta/\tau_0)\ (dp/dx)$. Stated differently, all of the members of the equilibrium family of laminar layers with rising gradients are obtained with practically the same pressure distribution, p vs. x, except those for which the gradients are relatively mild. Let us see what the effect of this condition is on downstream stability. Suppose that an equilibrium laminar layer is disturbed so that in one region θ, say, is increased, $(\theta/\tau_0)\ (dp/dx)$ will be increased, but $(x/q)\ (dp/dx)$ will be unchanged (we suppose the boundary layer has no effect on the outer flow). Now if the boundary layer originally corresponded to a point near the origin of the curve of Fig. 26, the disturbance will shift the point to the right of the curve. Any local irregularities in the profile will be ironed out by viscosity. In addition, there will be a progressive change in the gross characteristics of the profiles because they no longer lie on the equilibrium curve of Fig. 26. Intuition and experience say that the profiles will asymptotically return to a point on this curve.

Now suppose the original layer corresponded to a point on the flat part of the curve of Fig. 27. Again the disturbed point moves to the right, but there will be no tendency to return because it is still on the equilibrium curve. Such a layer would not possess downstream stability.

Let us see what happens in the case of turbulent profiles. As shown earlier, our family of turbulent profiles is not expressible in terms of θ, which is related to the universal length parameter Δ through the skin-friction coefficient c_f. However, since the Reynolds Number effect is not great for turbulent profiles, we may establish upper and lower limits for the relation between $(x/q)\ (dp/dx)$ and $(\theta/\tau_0)\ (dp/dx)$ if we are willing to restrict the range of Reynolds Numbers. We have

$$\frac{x}{q}\frac{dp}{dx} = \frac{(\theta/\tau_0)\ (dp/dx)}{\theta/xc_f}$$

The momentum equation,

$$\frac{d\theta}{dx} = \frac{c_f}{2} + \frac{(H+2)}{2}\frac{\theta}{q}\frac{dp}{dx} = c_f\left[1 + (H+2)\frac{\theta}{\tau_0}\frac{dp}{dx}\right]$$

with appropriately averaged c_f (denoted $\bar{c}_f$) can be integrated to give

$$\frac{\theta}{x\bar{c}_f} = \frac{1}{2}\left[1 + (H+2)\frac{\theta}{\tau_0}\frac{dp}{dx}\right]$$

Substituting this in the equation above, we have

$$\frac{x}{q}\frac{dp}{dx} = \frac{2(\theta/\tau_0)\ (dp/dx)}{1 + (H+2)\ (\theta/\tau_0)\ (dp/dx)}$$

This equation with upper and lower limits on the variation of c_f (and on H, which also varies with C_f) has been plotted on Fig. 26, using values obtained from our experimental profiles. This band has a number of interesting features. It shows that turbulent boundary layers can withstand about three times the nondimensional pressure gradient $(x/q)\ (dp/dx)$ that laminar profiles can. The greatest value for this parameter for equilibrium profiles is about 0.58. This means that a flow with

$$\frac{x}{q}\frac{dp}{dx} = -\frac{2x}{U}\frac{dU}{dx} = 0.58$$

or $U = \text{const.}\ x^{-0.29}$ can be decelerated to zero velocity without separation, but this is the maximum rate of deceleration which can be accomplished without eventual separation.

In Fig. 26, it is also seen that the turbulent boundary layer behaves similarly to the laminar boundary layer in that $(x/q)\ (dp/dx)$ is proportional to $(\theta/\tau_0)\ (dp/dx)$ for small pressure gradients, but the curve quickly levels off for values of $(\theta/\tau_0)\ (dp/dx)$ between 1 and 5. Above this point, a surprising thing happens. The dimensionless pressure gradient $(x/q)\ (dp/dx)$ actually decreases for profiles closer to separation. The peak values of $(x/q)\ (dp/dx)$ lie between 0.5 and 0.58. As $(\theta/\tau_0)\ (dp/dx)$ approaches ∞, corresponding to separation, $(x/q)\ (dp/dx)$ approaches $2/(H+2)$. At separation, however, H lies between 2.2 and 2.6.[8] The corresponding values of $(x/q)\ (dp/dx)$ at separation are 0.43 and 0.48, showing a definite decrease from the peak values. At first sight this behavior seems to be entirely incorrect, for it implies that the pressure distributions producing near-separating profiles are definitely flatter than those producing profiles well removed from separation. Further, since the curve has a maximum, this says that there are cases where two quite different boundary layers will be produced by exactly the same pressure distribution. Both will be shape-preserving, but one will have a shape only slightly different from the familiar constant

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pressure profile while the other will be very near separation. A typical example is $(x/q)\ (dp/dx) = 0.48$, which can be integrated to give $U = \text{const}\ x^{-0.24}$. In such a velocity distribution, Fig. 26 shows that an equilibrium boundary layer can exist for which $(\theta/\tau_0)\ (dp/dx) = 1.2$ with $H = 1.3$, a value very near that for constant pressure profiles. In exactly this same velocity distribution, a second layer can exist for which $(\theta/\tau_0)\ (dp/dx) = 15.0$ and $H = 2.1$, which is near the separation limit of $2.2 < H < 2.6$. This shows that two layers in the same pressure distribution can be at opposite extremes of the family for adverse pressure gradients.

This unexpected behavior can perhaps best be explained by examining the identity

$$\frac{x}{q}\frac{dp}{dx} \equiv \frac{(\theta/\tau_0)\ (dp/dx)}{\theta/xc_f}$$

When the pressure gradient is small, the rate of growth of θ with x is governed primarily by c_f, and if x is measured from the apparent origin of the boundary layer, θ/xc_f will be nearly constant. For this condition, the formula shows that $(x/q)\ (dp/dx)$ will be nearly proportional to $(\theta/\tau_0)\ (dp/dx)$, as one would intuitively think it should. However, as the pressure gradient increases, the pressure gradient quickly begins to dominate the small skin-friction term in the momentum equation. Under this condition, θ/xc_f is more nearly proportional to $[(H + 2)/2]\ (\theta/\tau_0)\ (dp/dx)$, and $(x/q)\ (dp/dx)$ becomes nearly equal to $2/(H + 2)$. If H were constant, $(x/q)\ (dp/dx)$ would be constant. However, H increases with $(\theta/\tau_0)\ (dp/dx)$, causing $(x/q)\ (dp/dx)$ to *decrease* for large $(\theta/\tau_0)\ (dp/dx)$, explaining the observed shape of the curve. The above discussion shows that, for values of $(\theta/\tau_0)\ (dp/dx)$ greater than 1 to 1.5, both laminar and turbulent layers will have little downstream stability and turbulent layers can show instability.

Our experience in obtaining our second set of equilibrium profiles confirms the above analysis. It will be remembered that these profiles were closer to separation. The $(\theta/\tau_0)\ (dp/dx)$ was approximately 5. Whenever a change in conditions was made, the effect of the change was felt with increasing amplitude downstream. However, the rate of divergence was not unmanagably large. Once the phenomenon was understood, the method of obtaining equilibrium profiles was changed from a process of simply correcting the pressure distribution to a process of "steering" the boundary layer along a G = constant course. Experimentally, the experience greatly resembled the process of driving a mildly unstable automobile that always wants to wander off the road and must be brought back. Once a satisfactory equilibrium layer had been obtained, it showed little tendency to wander off with time. The downstream instability is an instability in x and not in t.

Difficulties Obtaining Two-Dimensional Flow

During the course of our investigation, considerable difficulty was experienced obtaining two-dimensional flow. It is felt that these results have considerable influence on the question raised recently by several authors about the proper form for the momentum equation.

Wieghardt[10] and Mangler[11] conducted experiments on turbulent boundary layers in rising pressure gradients. They used the two-dimensional momentum equation to deduce the values of c_f. Their results indicate that c_f, instead of decreasing as separation was approached, actually increases. Schubauer[5] and Ludwieg,[3] using direct measurements of shear, found that c_f decreases in rising pressures. However, when the two-dimensional momentum equation is used to obtain c_f from Schubauer's data, rising values of c_f are obtained. Ludwieg has suggested that lack of two-dimensionality is the cause for the discrepancy. On the other hand, Newman,[12] Goldschmied,[13] and Bidwell[14] suggest that the discrepancy is caused by the momentum equation being in error through the omission of terms containing the turbulent fluctuations.

During the course of the present experiments, it was found that agreement with the two-dimensional momentum equation was obtained when the pressure was constant, but when the pressure was rising, definite disagreement was obtained. For our second rising pressure distribution, with near-separating profiles, the disagreement was very great. The skin-friction term was negligible compared with the $d\theta/dx$ and $[(H + 2)/2]\ (\theta/q)\ (dp/dx)$ terms. Further, the $[(H + 2)/2]\ (\theta/q)\ (dp/dx)$ term was 50 to 100 per cent greater than the $d\theta/dx$ term. If c_f had been computed from the momentum equation, a large negative value would have been obtained.

An attempt was made to obtain agreement with the momentum equation by taking into account the fluctuation terms as suggested by Newman, Goldschmied, and Bidwell. However, this procedure led to corrections far too small to explain the observed discrepancies. In the case of equilibrium profiles, with similar conditions existing at each station, the fluctuation terms in the momentum equation are either zero or very small. The reason for this is that these terms usually occur as an x derivative of an integral of the fluctuations. With similarity in conditions, these integrals are substantially constant.

An investigation was made to see if the discrepancy was caused by nonlinearity in the instrumentation, which would give incorrect mean velocity measurements in the presence of turbulent fluctuations. Mean velocities were measured by flattened hypodermic probes and by hot-wires. Turbulence measurements were also made by hot-wires. The hypodermic probes were calibrated against angles of attack and yaw. Knowing the levels of the u fluctuations and estimating

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the levels of the v and w fluctuations, estimates were made of the effect of the turbulence on the mean velocity readings of both the hypodermic probes and the hot-wires. Reasonable agreement was obtained except near the wall where the turbulence levels were relatively large. It was found that the corrections had insufficient effect on the integrated values of θ and δ^* to explain the momentum discrepancy.

From the above it was concluded that neither the additional terms proposed by Newman, Goldschmied, and Bidwell nor instrumentation errors could account for observed disagreement with the momentum equation. The evidence pointed to the possibility that the flow was not two-dimensional. Apparently, the flow was spreading laterally in such a way as to tend to thin out the boundary layer. If w is the local component of lateral velocity parallel to the plate (w is assumed constant with y), then a term $(\delta^*/U)\ (\partial w/\partial z)$ must be included in the momentum equation. Calculations showed that the lateral flow required to explain the discrepancy appearing in the momentum equation was small. At the sides of the tunnel, a divergence angle of about $1^1/_2{}^\circ$ would have been sufficient. Attempts were made with yawmeters to measure the lateral flow, but these were unsuccessful because the lateral turbulent fluctuations were so large (greater than $\pm$ 10° in flow inclination) that they masked any small shift in mean flow direction.

It will be remembered that the tunnel is constructed so that the boundary layers on the side walls are removed. It was conjectured that this removal might be causing the flow to diverge laterally. The slots were taped up, and the discrepancy was reduced but not eliminated. This was surprising because now the wall boundary layers should have caused the flow on the test floor to converge and reverse the discrepancy. Next, it was noticed that a crack, about 1/32 to 1/16 in. wide, existed where the walls joined the floor. This was taped up, and the discrepancy was again improved but not eliminated or reversed. At one section of the tunnel, this last change caused a previously well-behaved boundary layer on the floor to separate. This indicates that these fine cracks had been removing sufficient low-velocity air to prevent separation. An analysis of this situation showed that, when a leak such as these cracks exists, there is a strong tendency for the swiftly moving central stream to travel straight down the tunnel but for the slowly moving air next to the floor and walls to move laterally toward the leak.

Next we considered the possibility that the boundary layer air on the floor was moving laterally to the walls and then up through the wall boundary layers to the outlet flaps on the top of the tunnel, which were used to control the pressure distribution. Measurements of the wall boundary layers indicated they were not growing, even though the wall slots were closed. This strengthened our belief that low-velocity air was moving through the boundary layers to the outlet flaps. To prevent this movement, the ends of the outlet flap openings next to the walls were closed. This was finally sufficient to reverse the discrepancy. Once the discrepancy was under control, it was an easy matter to adjust conditions so that the two-dimensional momentum equation was satisfied.

After working to cure this discrepancy as long as we did, we came to have a great respect for the ease with which air can move laterally in boundary layers subjected to adverse pressures gradients. None of this trouble appeared when the pressure was constant. After having had this experience, it is easy to believe that the results of Mangler, Wieghardt, and Schubauer should be interpreted as indicating that their flows were not two-dimensional.

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